YICHONG CHEN

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RESEARCH INTERESTS

Edge AI, Distributed Computing, Adaptive Neural Networks, Deep Learning

EDUCATION BACKGROUND

Imperial College London *Apr 2022 - present*
Major: Computing
Degree: Ph.D
Research: AI/ML-driven system management for dynamic distributed edge computing environments.

Imperial College London *Sep 2019 - Nov 2020*
Major: Computing (Artificial Intelligence and Machine Learning)
Awarded: Distinction (80.2%)
Degree: Master of Science
Courses: Reinforcement Learning, Deep Learning, Natural Language Processing, Computer Vision.

University of Birmingham *Sep 2017 - Jun 2019*
Major: Mathematics
Awarded: First Class Honor (81%)
Degree: Bachelor of Science
Courses: Applied Statistics, Non-linear Programming & Heuristic Search, Real and Complex Analysis

South China University of Technology *Sep 2014 - Jun 2019*
Major: Information Management and Information System
Degree: Bachelor of Engineering
Courses: Data Mining and Statistics Decision, Object-oriented Programme Design, Probability, Operational Research

RESEARCH EXPERIENCE

Ph.D. Researcher, Imperial College London *Apr 2022 – Present*

- Developed a collaborative AI system with early-exit neural networks and predictive models, enabling faster and more energy-efficient inference on edge devices without sacrificing accuracy.
- Enhanced the AI system with reinforcement learning and automatic change detection to maintain high performance even when network conditions and workloads shift unexpectedly.
- Built a dynamic scheduler using simulation and online traffic prediction, significantly improving task allocation and reducing service delays and failures under real-world conditions.

PROFESSIONAL EXPERIENCE

Full Stack Developer, Imperial Consultants *Sep 2022 – Oct 2024*

- Led the design, development, and cloud deployment of SiMON, a solar anomaly detection platform used by SEENG LTD to monitor large-scale solar fields.
- Delivered a complete web-based application integrating client-provided detection algorithms, enabling real-time monitoring and field diagnostics.
- Supported SEENG LTD in securing new funding and external contracts by providing a robust, production-ready system that demonstrated clear business value.

AI Engineer, Huawei Technologies Co., Ltd *Nov 2020 - Jun 2022*

- Built and deployed AI systems for real-time monitoring of factory data, enabling early detection of quality issues in production lines.
- Developed interactive reporting tools to automate failure analysis and identify root causes, streamlining engineering workflows.
- Applied NLP models to extract insights from technical logs and built a searchable knowledge graph to support intelligent queries and decision-making.

PUBLICATIONS

Y. Chen, Z Niu, M Roveri, G Casale. CEED: Collaborative Early Exit Neural Network Inference at the Edge, in *Proc. of INFOCOM*, May 2025.

S. Wang, K Li, D You, **Y Chen**, A Lin, S Liu, X Li, JA McCann. Wimans: A benchmark dataset for wifi-based multi-user activity sensing , in *Proc. of ECCV*, Oct 2024.

Y. Chen, M Roveri, S Tuli, G Casale. Coupling QoS Co-Simulation with Online Adaptive Arrival Forecasting, in *Proc. of IFIP/IEEE CNSM*, Nov 2023.

Y. Chen, G. Casale. Deep Learning Models for Automated Identification of Scheduling Policies, in *Proc. of IEEE MASCOTS*, Nov 2021.

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, JavaScript
Frameworks & Tools	PyTorch, Flask, React, Vue3
Languages	English (Business Conversation), Mandarin (Native), Cantonese (Native)
Website	https://chanyikchong.github.io/